

AGFG2 (O95081) — ArfGAP Catalysis vs. Trafficking Cofactor

Focus type: function_assignment **Hypothesis slug:** arfgap-catalysis-versus-trafficking-cofactor

Question: Does human AGFG2/HRBL directly possess GTPase-activator (ArfGAP) activity, warranting a molecular-function annotation, or is its established role a non-catalytic vesicle-trafficking cofactor?

Executive Judgment

Verdict: UNRESOLVED (leaning "unsupported as an experimental MF; not refuted at the sequence level").

There is **no direct biochemical evidence** that AGFG2 accelerates GTP hydrolysis on any ARF (or other) GTPase. The molecular-function claim of catalytic ArfGAP activity therefore cannot be asserted as experimentally supported. At the same time, it **cannot be refuted on sequence grounds**: AGFG2 retains an intact Arf-GAP zinc-finger domain and, critically, the catalytic **arginine finger (Arg75)** that the transition-state structure of ASAP3–Arf6 identifies as the key catalytic residue (PMID20510928). The one divergence highlighted by the seed — **Thr89** where ASAP3 has **Asp484** — is not, by itself, diagnostic of inactivity, because the GIT-family ArfGAPs lack that same aspartate-homologous site (PMID23433073) yet display measured ARF GAP activity (PMID10788515).

The only functional data on AGFG2 (PMID34369554) is a knockdown secretion phenotype (von Willebrand factor exocytosis from Weibel–Palade bodies), which is **not a GTPase-hydrolysis assay**. The "GAP-dead" R75Q construct in that paper still promoted secretion, but those data are marked "not shown." Thus the evidence-supported role of AGFG2 is a **trafficking cofactor**, while catalytic ArfGAP activity remains an untested computational inference. Absent assays are not equivalent to functional refutation.

Most important caveat: the central factual question — has anyone ever purified AGFG2 (or its paralog AGFG1) and measured GAP activity? — appears to be answered "no" in the retrievable literature. The MF annotation therefore rests on domain homology (IEA/ISS), not on experiment.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ limitations
PMID20510928 (Ismail 2010)	Structural (transition-state X-ray)	Qualifies	Which residues define the ArfGAP catalytic mechanism	Arginine finger (ASAP3 Arg469) + Arf6 glutamine + a Ca ²⁺ ion are the key catalytic residues; the Asp position (=AGFG2 Thr89) is not listed among essential catalytic residues	ASAP3–Arf6 fusion, in vitro	High confidence; ASAP3 not tested; AGFG2 stimulates ASAP3
PMID10788515 (Vitale 2000)	Direct biochemical assay	Qualifies / supports feasibility	Is the Asp-homologous site required for catalysis?	GIT1/GIT2 show ARF6 GAP activity ; zinc finger required; 1 Zn ²⁺ /molecule	Recombinant human proteins, in vitro	High; precise; not Arf6 itself
PMID23433073 (Schlacht 2013)	Evolutionary / comparative	Qualifies	Catalytic-residue conservation across the ArfGAP family	AGFG domains show catalytic divergence (Asp and conserved Trp); the Asp-homologous site is lost in all 18 GIT sequences	Multi-eukaryote sequence phylogeny	High confidence; evolutionary prediction; not as active
PMID34369554 (Watanabe 2021)	Knockdown / mutant phenotype	Qualifies (not an MF assay)	Does AGFG2 function in secretion; is catalysis required?	AGFG2 KD inhibits stimulated vWF exocytosis; R75Q "GAP-dead" still promotes secretion (data not shown)	HUVEC/ endothelial, cell-based	Model secretion assay; hydrolysis assay; data not shown
This work	Computational alignment	Qualifies	Verify AGFG2↔ASAP3	AGFG2 Arg75↔ASAP3	UniProt O95081 (dom	

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	(NW/ BLOSUM62)		catalytic- residue mapping	Arg469 (arginine finger conserved); AGFG2 Thr89↔ASAP3 Asp484 ; C4 zinc- finger cysteines intact	27–153, ZnF 47–70) vs Q8TDY4 (dom 426– 551, ZnF 441–464)	Reproducible inference assay
This work	Computational alignment vs canonical active GAP	Qualifies	Cross-check arginine finger + tryptophan conservation	Reciprocal best match AGFG2 Arg75↔ARFGAP1 Arg50 (independent confirmation of arg finger); a Trp in the ARFGAP1 catalytic core does not align to a Trp in AGFG2 (consistent with PMID23433073 Trp divergence)	UniProt O95081 vs ARFGAP1 Q8N6T3 (dom 7–124)	Ragged region finger robust call suggested
UniProt O95081 / P52594	Database record	Qualifies	Provenance of existing GAP annotation	GO:0005096 is IEA:InterPro for both AGFG2 and AGFG1 (no experimental GAP); vWF/WPB secretion not yet in GO	Human, database	Datab level; confir homo only s

GO Curation Implications (leads — require curator verification)

Curation baseline (verified from UniProt O95081, iteration 2): The existing MF term GO:0005096 "GTPase activator activity" is annotated **IEA:InterPro** (from ArfGAP domain IPR001164 / PF01412) — i.e., pure domain-based inference, no experimental support. AGFG2's curated BP terms are all **IBA:GO_Central** (acrosome assembly, intermediate filament organization, spermatid nucleus differentiation); **the experimentally demonstrated regulated-exocytosis role (PMID34369554, vWF/Weibel-Palade body) is NOT yet in GO.** The paralog AGFG1 (P52594) also carries GO:0005096 only as **IEA:InterPro**, while its experimental annotations are nucleus / cytoplasmic vesicle / RNA binding / mRNA export — so *neither* AGFG paralog has an experimentally supported GAP annotation.

Molecular Function — "ARF GTPase activator activity" (GO:0008060) / "GTPase activator activity" (GO:0005096): - Recommended action: retain only as computational/IEA (or ISS) — **do NOT upgrade to an experimental (EXP/IDA) MF annotation.** No direct assay supports catalysis for AGFG2; the current term is already IEA:InterPro, which is appropriately conservative. - **Do not remove** the term solely on the Thr89 (Asp→Thr) substitution: the GIT precedent (PMID10788515 + PMID23433073) shows loss of that aspartate is not diagnostic of inactivity, and the catalytic arginine finger (Arg75) is retained. - If an experimental MF annotation currently exists based on PMID34369554, it is **too strong** — that paper provides a cellular phenotype, not GAP activity; consider re-scoping to a BP annotation.

Biological Process (evidence-supported, core): - "regulated exocytosis" / "von Willebrand factor secretion" / "Weibel-Palade body" (e.g., GO:0060627 regulation of vesicle-mediated transport; GO:0002576-adjacent secretion terms; GO:0033093 Weibel-Palade body) — supported by IMP/IGI from PMID34369554 (siRNA knockdown). This is the best-supported functional annotation.

Cellular Component: AGFG family carries FG-repeat / nucleoporin-related regions; localization annotations should follow direct evidence, not be inferred from the MF claim.

GO Decision Table (leads — require curator verification)

GO ID	Term	Current status	Recommended action (lead)	Rationale
GO:0005096	MF: GTPase activator activity	IEA:InterPro (verified)	Retain as IEA/ISS; do NOT upgrade to EXP	No direct GAP assay on AGFG2; zinc-finger + arginine finger intact but Thr89/Trp divergence — homology-only inference
GO:0008060	MF: ARF GTPase activator activity (more specific)	Not annotated	Optional ISS lead only	More specific, still unproven; substrate ARF isoform untested
GO:0032796 / regulated exocytosis (e.g. GO:0045055)	BP: regulated exocytosis / vWF secretion	Not in GO	ADD as IMP (PMID34369554)	Knockdown reduces stimulated vWF exocytosis — best-supported functional annotation
GO:0033093	CC: Weibel-Palade body (context)	Not in GO	Consider as context; needs direct localization	Functional link shown; direct AGFG2 localization to WPB not established

Structural provenance of the catalytic scaffold (iteration 3)

Direct inspection of UniProt sequences shows the AGFG2 ArfGAP zinc finger (res 47–70) has four cysteines at 47, 50, 67, 70 with inter-cysteine gaps [3, 17, 3] — **identical spacing to canonical active ARFGAP1** (Cys 22, 25, 42, 45; gaps [3, 17, 3]; the C-x2-C-x16-C-x2-C ArfGAP zinc knuckle). The catalytic arginine finger sits immediately C-terminal to the last cysteine in a conserved motif — AGFG2 SGLLRGL (R75), ARFGAP1 SGRHRGL (R50), ASAP3 SGVHREL (R469).

The zinc-coordinating scaffold and arginine finger of AGFG2 are therefore structurally canonical and intact, so the MF claim cannot be refuted on structural grounds; divergence is confined to accessory positions (Thr89↔Asp484 and a non-conserved tryptophan).

Bottom line for the curator: the ArfGAP MF term is best treated as **non-core / homology-only**, and the gene's evidence-backed core is a **trafficking cofactor in ARF-regulated regulated secretion**. Avoid defaulting to "protein binding."

Mechanistic Scope

- **Immediate molecular activity tested:** acceleration of GTP hydrolysis on an ARF substrate by the AGFG2 Arf-GAP zinc-finger domain (an MF, GO:0008060).
- **What is actually demonstrated:** a downstream **cellular phenotype** — loss of AGFG2 reduces stimulated vWF exocytosis (PMID34369554). This is separated from direct catalysis by at least two inferential steps (knockdown → pathway effect; not enzyme kinetics).
- **Structural inference:** catalysis in this family is driven by a GAP-supplied arginine finger acting on an Arf-supplied glutamine (PMID20510928); AGFG2 retains the arginine finger, so the catalytic machinery is not obviously broken, but this is inference, not measurement.

Conflicts and Alternatives

1. **Non-catalytic cofactor model (favored by available data).** The R75Q "GAP-dead" construct reportedly still promoting secretion (PMID34369554, data not shown) suggests the trafficking role may not require catalysis. Catalysis and cofactor roles are **not mutually exclusive**, but only the cofactor role has functional support.
2. **Paralog / family carry-over.** ArfGAP MF annotations propagate by InterPro domain (Arf-GAP, PF01412) across the whole family; AGFG2's term may be database carry-over rather than gene-specific evidence.
3. **Evolutionary divergence argument (PMID23433073).** Predicts AGFG catalytic weakening (Asp + Trp divergence). This is a prediction, self-limited by the authors' own GIT counterexample.
4. **Substrate ambiguity.** Even if AGFG2 is an active GAP, the target Arf isoform is unknown (GIT proteins prefer ARF6 over ARF1; AGFG2 substrate untested).

5. AGFG1 context. AGFG1/HRB was historically characterized as an HIV-Rev cofactor / FG-repeat nucleoporin-like protein, not as an assayed GAP — reinforcing that the family's "ArfGAP" label is largely domain-based.

Knowledge Gaps

Gap	What was checked	Why it matters	Resolving evidence
No purified-protein GAP assay for AGFG2	PubMed searches (AGFG2, WPB, GIT/ArfGAP) returned no biochemical assay	Determines whether MF term is experimentally justifiable	In vitro GAP assay (single-turnover GTP hydrolysis) on ARF1/ARF5/ARF6 ± AGFG2 domain
Catalytic role of the Asp position	Structure (PMID20510928) lists arg finger + Arf Gln, not Asp; GIT lacks Asp yet active	Decides whether Thr89 refutes activity	Asp→Thr / Thr→Asp swap assays in an active ArfGAP scaffold
Whether R75/ catalysis is needed for secretion	PMID34369554 states R75Q rescues, "data not shown"	Distinguishes catalytic vs scaffold mechanism	Published, quantified R75Q rescue with a validated GAP-dead control
Substrate identity	No AGFG2 substrate mapping found	Needed for a precise MF/substrate qualifier	GAP assay panel across ARF isoforms

Discriminating Tests

- 1. Single-turnover GAP assay** with recombinant AGFG2 Arf-GAP domain (res ~27–153) against ARF1/ARF5/ARF6 loaded with GTP; include ASAP3 (positive) and a catalytic-Cys-mutant AGFG2 (negative). Direct readout of the MF claim.
- 2. Structure-guided mutagenesis:** AGFG2 R75Q (arg-finger dead) and T89D (restore Asp) tested in the GAP assay to determine whether Thr89 limits activity.

3. **Rescue specificity in cells:** re-express WT vs R75Q vs zinc-finger-dead AGFG2 in AGFG2-depleted endothelial cells and quantify vWF exocytosis — resolves catalytic vs scaffold requirement.
 4. **Comparative alignment/AlphaFold3 modeling** of AGFG2–ARF6 to check whether the catalytic geometry (arg finger, Ca²⁺/Asp pocket) is preserved.
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Curation Leads (require curator verification)

- **Candidate references to attach:**

- PMID34369554 — AGFG2 in regulated vWF exocytosis (BP/IMP lead; note "GAP-dead" data not shown).
- PMID20510928 — defines ArfGAP catalytic residues (arg finger; Asp not essential) — supports NOT refuting on Thr89.
- PMID10788515 — GIT ArfGAP activity without the Asp-homologous site — counters over-reliance on the Asp substitution.
- PMID23433073 — evolutionary divergence of AGFG catalytic residues — context for weakening (but self-caveated).

- **Candidate GO actions:**

- MF GO:0008060 / GO:0005096 → keep as **IEA/ISS only; flag as unverified**; do not assign experimental evidence codes.
- BP → add/retain **regulated exocytosis / vWF secretion / Weibel-Palade body** with IMP from PMID34369554.

- **Suggested curator questions:**

- Is any existing experimental MF ArfGAP annotation on AGFG2 traceable to a real assay, or is it phenotype-based (PMID34369554)? If the latter, re-scope to BP.
 - Should the review add a "no direct catalytic assay available" note to prevent paralog-driven over-annotation?
 - **Suggested experiments:** the single-turnover GAP assay + R75Q/T89D mutants above.
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Limitations

- Analysis is literature- and sequence-based; no wet-lab data were generated.
 - The pivotal R75Q rescue in PMID34369554 is "data not shown," so its interpretation is provisional.
 - The pairwise NW alignment reproduces the seed's MAFFT coordinates but is a two-sequence global alignment, not a full family MSA; conclusions about the arginine finger and Asp positions are robust because they are anchored on the ASAP3 crystal structure numbering.
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